SEMAGLUTIDE

Once weekly

Pancreatitis, retinopathy

CHEMICAL NAME Aib8,Arg34-GLP-1(7-37) with C18 diacid–γGlu–OEG–OEG side chain on Lys26 (acylated GLP-1 analogue)

CLASS Glucagon-like peptide-1 (GLP-1) receptor agonist (weekly or oral)

TRADE NAMES Ozempic (weekly SC) · Rybelsus (oral) · Wegovy (weight management)

THERAPEUTIC USES / INDICATIONS

- Type 2 diabetes (CV benefit; SUSTAIN-6).
- Chronic weight management (Wegovy, BMI ≥30 or ≥27 with comorbidity).
- Cardiovascular risk reduction in obesity (Wegovy).

ADMINISTRATION

- **SC once weekly** on the same day, any time, with or without meals.
- **Oral (Rybelsus):** on an empty stomach with ≤120 mL water, then wait 30 min before food, drink or other drugs.
- Missed weekly dose: give within 5 days; otherwise skip.

DOSAGES

- **Ozempic:** 0.25 mg weekly for 4 weeks (titration), then 0.5 mg; may increase to 1 mg, then 2 mg.
- **Wegovy:** 0.25 mg weekly, escalate every 4 weeks to 2.4 mg.
- **Rybelsus:** 3 mg daily for 30 days, then 7 mg; may increase to 14 mg.

PHARMACOKINETICS

- **Peak:** SC 1–3 days. **t½:** ~1 week (once-weekly dosing).
- Albumin binding; proteolytic degradation.
- Excreted in urine and faeces.

PHARMACODYNAMICS

- **GLP-1 receptor** agonist → glucose-dependent insulin release, ↓ glucagon, slowed gastric emptying, strong appetite suppression → significant weight loss (~10–15%).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, vomiting, diarrhoea, constipation.
- Abdominal pain, dyspepsia, burping.
- Fatigue.

ADVERSE EFFECTS (SERIOUS)

- Acute pancreatitis; gallbladder disease.
- Diabetic retinopathy complications (rapid glucose fall).
- Dehydration → AKI.
- Hypoglycaemia with SU/insulin; aspiration risk under anaesthesia (retained gastric contents).

CONTRAINDICATIONS & CAUTIONS

- Personal/family history of medullary thyroid carcinoma or MEN2.
- Pregnancy (stop ≥2 months before conception), breastfeeding.
- Type 1 diabetes; history of pancreatitis (caution).

KEY INTERACTIONS

- **Insulin, sulfonylureas:** hypoglycaemia.
- **Warfarin:** monitor INR.
- **Oral levothyroxine (with Rybelsus):** ↑ exposure.

NURSING MANAGEMENT

- Teach weekly pen use and dose escalation.
- Monitor weight, HbA1c, GI tolerance, hydration.
- Eye screening in diabetic retinopathy.
- Inform anaesthetist pre-operatively (aspiration risk).

PATIENT EDUCATION

- Inject once a week on the same day.
- Eat smaller meals; nausea usually settles.
- Report severe abdominal pain, vision changes, persistent vomiting.
- Weight regain is common after stopping.

KEY POINTS

- Weekly GLP-1 agonist (also oral).
- Strong weight loss.
- Titrate slowly to reduce nausea.
- Pancreatitis, retinopathy, gallstones.